FairShare - Project Documentation

1. Project Description

FairShare is a group expense-sharing application designed to simplify tracking, splitting, and settling shared expenses among friends, roommates, or teams. Users can create groups, log expenses, view individual contributions, and track balances transparently. The application provides notifications, cloud-based data storage, and real-time updates.

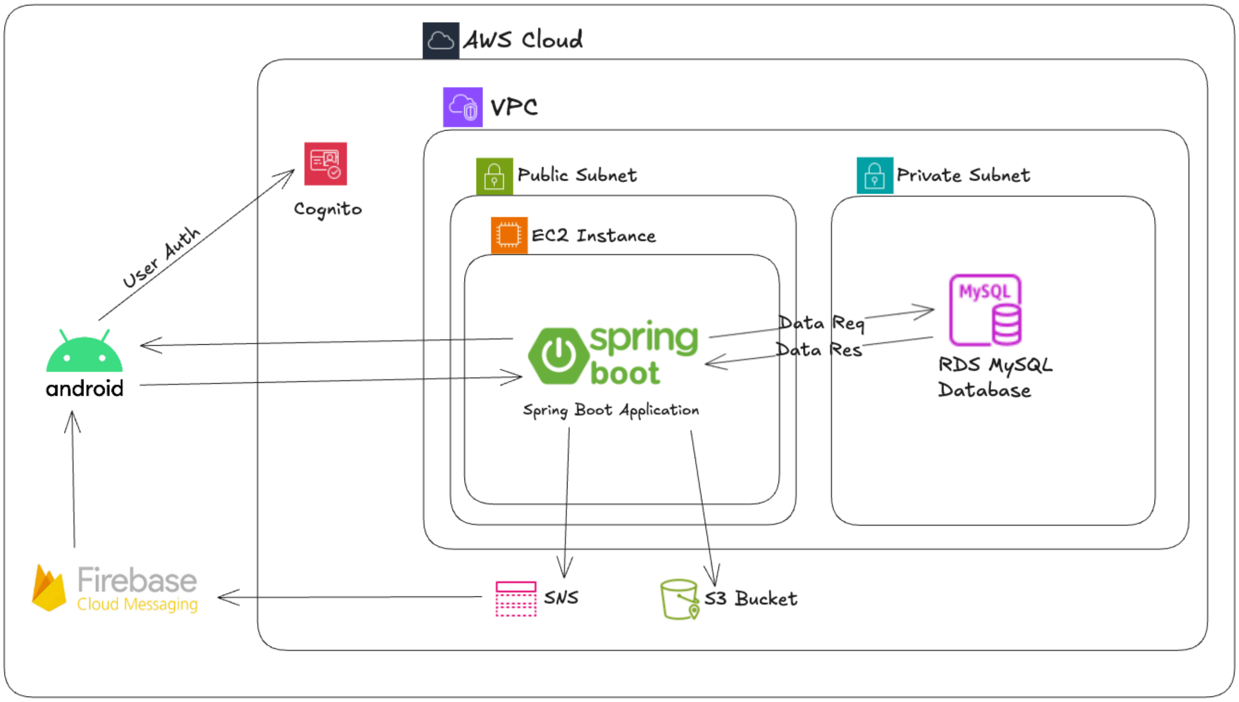
The backend is developed using Spring Boot, deployed on AWS EC2, and uses AWS RDS (MySQL) for data persistence. The Android app is the primary frontend, and AWS SNS/Firebase is used for push notifications.

2. Features & Functionality

Core Features:

- **Group Management:** Create, join, or leave expense-sharing groups.
- **Expense Logging:** Add new expenses, split between members, and assign payers.
- **Balance Summary:** Shows how much each member owes or is owed.
- **User Authentication:** Simple email-based authentication.
- **Push Notifications**: Receive real-time alerts when new expenses are added.

3. Architecture Overview



4. Technologies Used

4.1 Frontend

- **Platform**: Android (written in Kotlin)
- **Tools & Libraries**:
  - **Jetpack Components**: ViewModel, LiveData, Navigation
  - **Retrofit**: Network layer to communicate with backend APIs
  - **Firebase Cloud Messaging (FCM)**: For receiving push notifications

- **Functionality**:
  - Group creation and member management
  - Adding and viewing shared expenses
  - Visual analytics (Pie chart for spending distribution)
  - Real-time sync with backend

4.2 Backend

- **Framework**: Spring Boot (Java)
- **Build Tool**: Maven
- **Database Access**: Spring Data JPA + Hibernate
- **APIs**: RESTful endpoints consumed by the Android app
- **Push Notifications**: Integrated **AWS SNS** with **FCM** to send device-specific push notifications.

4.3 AWS Services

Amazon Cognito: Handles user authentication and authorization. Manages sign-up and login processes for the mobile application.

AWS Lambda: Triggers a post-confirmation function to automatically verify users after successful sign-up in Cognito.

Amazon S3: Stores uploaded expense receipt images. Provides secure access to these assets through object URLs.

Amazon SNS: Sends push notifications to user devices through integration with Firebase Cloud Messaging (FCM).

Amazon RDS: Hosts the MySQL database used by the backend application. Runs in a private subnet within a custom VPC.

Amazon EC2: Hosts the Spring Boot application. The .jar file is deployed to an Ubuntu instance.

Amazon VPC: Provides network isolation and configuration. Contains public and private subnets used by EC2 and RDS instances.

Security Groups: Manage network-level access (e.g., port 8080 open for API, MySQL access only from EC2).

IAM Roles: Used to manage access and permissions securely within AWS infrastructure.

5. App Installation & Test Login

Installing the Android App

To install the FairShare mobile app:

1. Locate the file fairshare-app.apk in the project zip folder.
2. Transfer the APK to your Android device using USB, Bluetooth, or any file-sharing method.
3. On your Android device:
    - Enable **"Install from unknown sources"** in Settings > Security.
    - Tap on the APK file to begin installation.
    - Follow the prompts to install the app.

Note: You may need to allow permission to install from your file manager or browser.

Test Login Credentials

To explore the app's functionality without signing up, you can use one of the following test account:

- **Emails:**
    - alice@test.com
    - bob@test.com

    - charlie@test.com

- **Password:** Password123

This account provides access to pre-created groups and sample expenses to help you test the app’s core features.